



Atomic Physics

原子物理学

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1 Rutherford 模型

Coulomb 散射

$$b = \frac{a}{2} \cos \frac{\theta}{2} \quad \left(a = r_{\min} = \frac{Z_1 Z_2 e^2}{4\pi\epsilon_0 E} \right)$$

Rutherford 散射

$$\sigma_c(\theta) = \frac{a^2}{16} \sin^{-4} \frac{\theta}{2} \quad N(\theta) \propto \sin^{-4} \frac{\theta}{2}$$

散射粒子数

$$N(\theta_1 \sim \theta_2) = N_0 n t \pi [b(\theta_2)^2 - b(\theta_1)^2] = N_0 \frac{\rho N_A}{M} t \pi [b(\theta_2)^2 - b(\theta_1)^2]$$

原子相关半径数量级

原子 $\sim 10^{-10} \text{ m}(\text{\AA})$, 原子核 $\sim 10^{-14} \text{ m}$, 电子 $\sim 10^{-18} \text{ m}$.

2 Bohr 理论

Stefan-Boltzmann 定律

$$R(t) = \sigma T^4$$

Wien 位移定律

$$\lambda_{\max} T = b$$

光电效应

$$E_k = h\nu = \phi = eU_a = e(\kappa\nu - U_0)$$

$$\Rightarrow \begin{cases} U_0 = \frac{\phi}{e} & \text{遏制电压} \\ h = e\kappa & \text{Planck 常量} \end{cases}$$

Rydberg 公式

$$\tilde{\nu}_H = R_H \left(\frac{1}{m^2} - \frac{1}{n^2} \right) = T(m) - T(n)$$

$$h\nu = hc\tilde{\nu} = \frac{hc}{\lambda} = E_n - E_m = hc[T(m) - T(n)]$$

氢原子与类氢离子的半径、速度、能级

$$r_n = \frac{n^2}{Z} a_0 \quad v_n = \frac{Z}{n} \alpha c \quad E_n = -\frac{Z^2}{n^2} \times 13.6 \text{ eV}$$

选择定则

$$\Delta l = \pm 1$$

3 量子力学基本原理

定态 Schrödinger 方程

$$\left(-\frac{\hbar^2}{2m}\nabla^2 + V\right)\psi = E\psi$$

一维无限深势阱

$$\psi(x) = \sqrt{\frac{2}{a}} \sin \frac{n\pi x}{a} \quad E_n = \frac{n^2 \hbar^2}{8ma^2}$$

一维方势垒贯穿概率

$$P = e^{-2\sqrt{2m(V-E)}\Delta x/\hbar}$$

动量算符

$$\hat{p}_x \rightarrow -i\hbar \frac{\partial}{\partial x}$$

力学量的平均值

$$\langle A \rangle = \int_{-\infty}^{\infty} \psi^* \hat{A} \psi dx$$

Hermite 算符的证明

$$\int_{-\infty}^{\infty} \phi^* (\hat{F} \psi) dx = \int_{-\infty}^{\infty} (\hat{F} \phi)^* \psi dx$$

对易关系

$$\begin{aligned} [\hat{x}_i, \hat{x}_j] &= [\hat{p}_i, \hat{p}_j] = 0 & [\hat{x}_i, \hat{p}_j] &= i\hbar \delta_{ij} \\ \hat{L}_i &= \varepsilon_{ijk} \hat{x}_j \hat{p}_k & [\hat{L}_i, \hat{v}_j] &= i\hbar \varepsilon_{ijk} \hat{v}_k & [\hat{L}_i, \hat{L}^2] &= 0 \end{aligned}$$

4 自旋与原子精细结构

总角动量

$$J = \sqrt{j(j+1)}\hbar \quad J_z = m_j \hbar \quad j = |l-s|, \dots, l+s$$

Landé g 因子

$$g = \frac{3}{2} + \frac{1}{2} \frac{s(s+1) - l(l+2)}{j(j+1)}$$

磁矩

$$\mu_j = -g_j \sqrt{j(j+1)} \mu_B \quad \mu_{j,z} = -g_j m_j \mu_B$$

$$\boldsymbol{\mu}_j = \boldsymbol{\mu}_L \cos(\mathbf{L}, \mathbf{J}) + \boldsymbol{\mu}_S \cos(\mathbf{S}, \mathbf{J})$$

附加能量

$$U = -m_j g_j \mu_B B$$

Zeeman 效应

$$\Delta E = (m_2 g_2 - m_1 g_1) \mu_B B$$

$$\Delta \nu = (m_2 g_2 - m_1 g_1) \mathcal{L} \quad \mathcal{L} = \frac{\mu_B B}{h}$$

$$\Delta \tilde{\nu} = (m_2 g_2 - m_1 g_1) \tilde{\mathcal{L}} \quad \tilde{\mathcal{L}} = \frac{\mu_B B}{hc}$$

正常 Zeeman 效应谱线的偏振

$$\Delta m_j = \begin{cases} +1 & \sigma^+ \\ 0 & \pi \\ -1 & \sigma^- \end{cases}$$

选择定则

$$\Delta S = 0 \quad \Delta L = \pm 1 \quad \Delta J = 0, \pm 1 \quad (0 \rightarrow 0 \text{ forbidden})$$

S-G 实验

$$Z_2 = -m_j g_j \mu_B \frac{\partial B_z}{\partial z} \frac{dD}{2E_k}$$

P-B 效应

$$\Delta(h\nu) = (2m_s + m_l) \mu_B B \quad \begin{cases} \Delta m_s = 0 \\ \Delta m_l = 0, \pm 1 \end{cases}$$

5 多电子原子

原子基态

1. $S_{\max}$;
2. $L_{\max}$;
3. 未半满/半满 $J_{\min}$, 过半满 $J_{\max}$.

6 X 射线

连续谱最小波长

$$\lambda_{\min} = \frac{hc}{eV} = \frac{1.24}{V(\text{kV})} \text{nm} \quad E_k = \frac{hc}{\lambda_{\min}}$$

Moseley 定律

$$\nu_{K_\alpha} = (Z - 1)^2 \times 0.246 \times 10^{16} \text{ Hz}$$

$$\Delta E_{K_\alpha} = (Z - 1)^2 \times 13.6 \text{ eV} \times \left(1 - \frac{1}{4}\right) \quad \Delta E_{K\text{-ionization}} = (Z - 1)^2 \times 13.6 \text{ eV}$$

Bragg 衍射

$$n\lambda = 2d \sin \theta$$

Compton 散射

$$\Delta\lambda = \lambda_C(1 - \cos \theta) \quad \lambda_C = \frac{h}{m_e c}$$

$$\Delta\lambda = \lambda_C \quad (\theta = 90^\circ)$$

$$h\nu' = \frac{h\nu}{1 + \gamma(1 - \cos \theta)} \quad \gamma = \frac{h\nu}{m_e c^2}$$

$$E_k = h\nu - h\nu' = \frac{hc}{\lambda} - \frac{hc}{\lambda'} = h\nu \frac{\gamma(1 - \cos \theta)}{1 + \gamma(1 - \cos \theta)}$$

$$(h\nu')_{\min} = \frac{h\nu}{1 + 2\gamma} \quad E_{k,\max} = h\nu \frac{2\gamma}{1 + 2\gamma} \quad (\theta = 180^\circ)$$

X 射线的吸收

$$I(x) = I_0 e^{-\mu x}$$